7.4 U-Net

- the main idea of FCNs is to supplement a usual contracting network by successive layers, where pooling operators are replaced by upsampling operators
- hence, these layers increase the resolution of the output
- in order to localize, high-resolution features from the contracting path are combined with the upsampled output
- a successive convolution layer can then learn to assemble a more precise output based on this information
- one important modification in the U-Net architecture is that in the upsampling part we have also a large number of feature channels, which allow the network to propagate context information to higher-resolution layers
- as a consequence, the expansive path is more or less symmetric to the contracting path, and yields a U-shaped architecture
- the network does not have any fully connected layers and only uses the valid part of each convolution, i.e., the segmentation map only contains the pixels for which the full context is available in the input image
- this strategy allows the seamless segmentation of arbitrarily large images

7.4 U-Net

- the network architecture is illustrated in Figure 8
- it consists of a contracting path (left side) and an expansive path (right side)
- the contracting path follows the typical architecture of a convolutional network
- it consists of the repeated application of two 3×3 convolutions (unpadded convolutions), each followed by a rectified linear unit (ReLU) and a 2×2 max-pooling operation with stride 2 for downsampling
- at each downsampling step, we double the number of feature channels
- every step in the expansive path consists of an upsampling of the feature map followed by a 2×2 transposed convolution ("up-convolution") that halves the number of feature channels, a concatenation with the correspondingly cropped feature map from the contracting path (skip connection, similar to ResNet), and two 3×3 convolutions, each followed by a ReLU
- the cropping is necessary due to the loss of border pixels in every convolution
- at the final layer a 1×1 convolution is used to map each 64-component feature vector to the desired number of classes
- in total, the network has 23 convolutional layers

7.4 U-Net

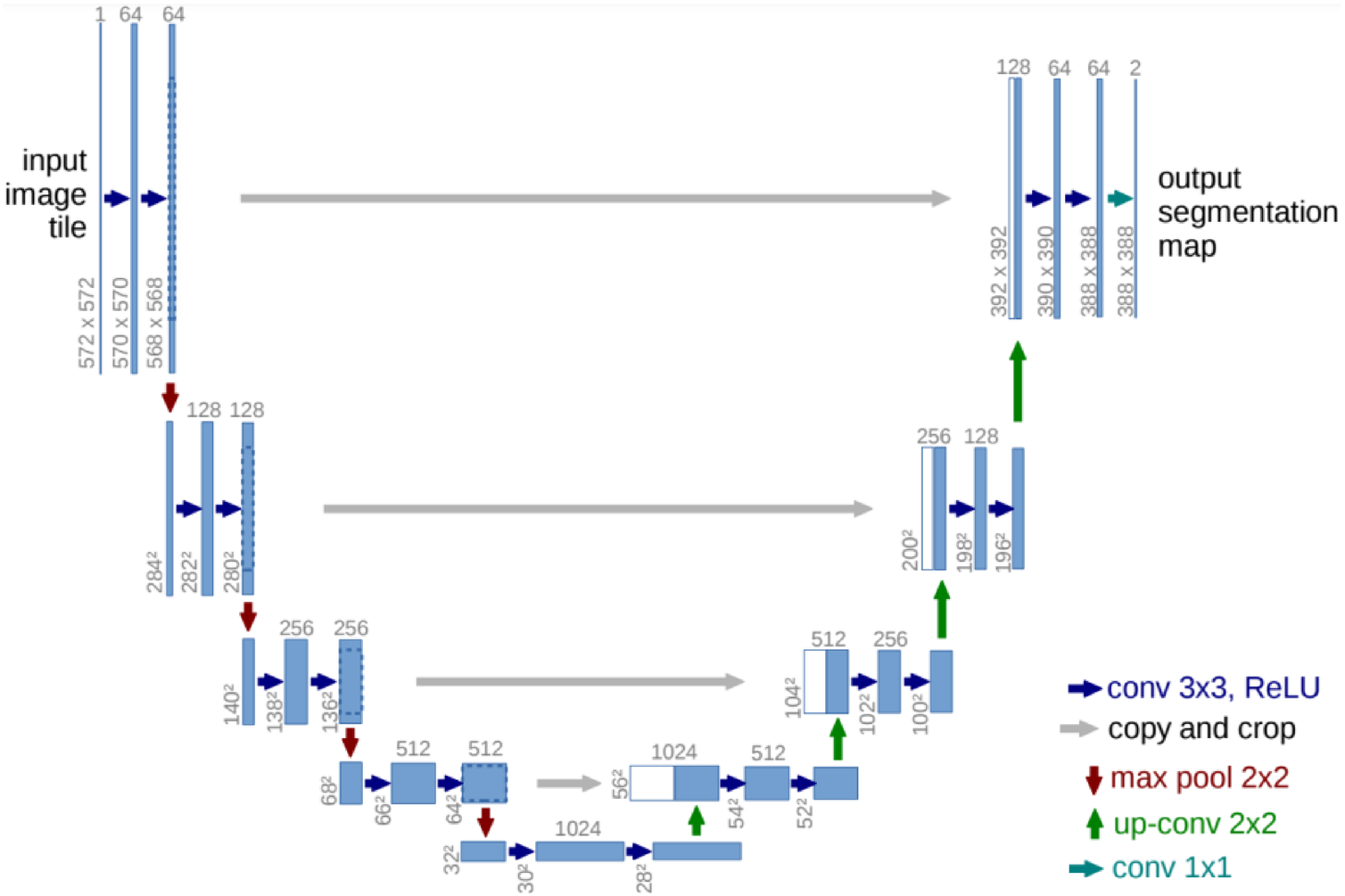


Figure 8: U-Net architecture. Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The *x*-*y* size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.